The KKT conditions and what they mean

min: $f_0(x)$

st: $f_i(x) \le 0$ $\quad \lambda_i \ge 0$

$h_j(x) = 0$ $\quad \nu_j$

$(\hat{x}, (\hat{\lambda}, \hat{\nu}))$ satisfy KKT if:

- $\hat{x}$ feasible
- $(\hat{\lambda}, \hat{\nu})$ dual feasible
- $\hat{\lambda}_i f_i(\hat{x}) = 0$ CS
- $\nabla f_0(\hat{x}) + \sum \lambda_i \nabla f_i(\hat{x}) + \sum \nu_j \nabla h_j(\hat{x}) = 0$ (*)

"$-\nabla f_0(\hat{x})$ makes $\ge 90^\circ$ with all allowable dir. from $\hat{x}$"

$\Updownarrow$

"any movement away from $\hat{x}$ will increase $f_0(\hat{x})$"

Why KKT is Useful

- KKT give a proof of optimality that is local.

- Also can be useful for computing $(\hat{x}, (\hat{\lambda}, \hat{\nu}))$.

Ex: $\min_x \; 3x^2 - x + 1$

$6x - 1 = 0$ $\quad x = 1/6$

An Example — 1/2

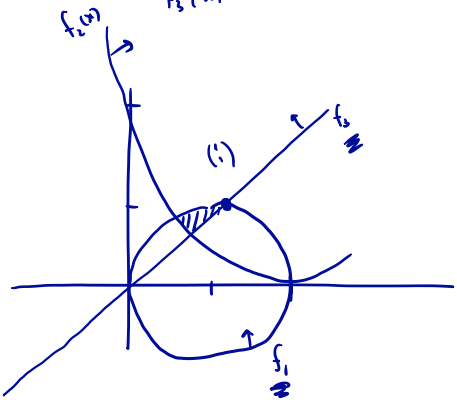
f_0:

min: $x_1^2 - 6x_1 + x_2^2 - 4x_2$

st: $f_1(x) = x_1^2 - 2x_1 + x_2^2 \le 0$

$f_2(x) = \frac{1}{2}x_1^2 - 2x_1 - x_2 + 2 \le 0$

$f_3(x) = x_1 - x_2 \le 0$



$$\nabla f_0(x) = \begin{pmatrix} 2x_1 - 6 \\ 2x_2 - 4 \end{pmatrix} //$$

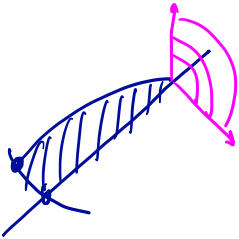
$$\nabla f_1(x) = \begin{pmatrix} 2x_1 - 2 \\ 2x_2 \end{pmatrix} \quad * \quad \underline{\underline{\begin{pmatrix} 0 \\ 2 \end{pmatrix}}}$$

$$\nabla f_2(x) = \begin{pmatrix} x_1 - 2 \\ -1 \end{pmatrix}$$

$$\nabla f_3(x) = \begin{pmatrix} 1 \\ -1 \end{pmatrix} \quad * \quad \begin{pmatrix} 1 \\ -1 \end{pmatrix}$$

An Example — 2/2

Claim: $\hat{x} = \begin{pmatrix}1\\1\end{pmatrix}$ is optimal.



$$\nabla f_0\Big|_{\binom{1}{1}} = \begin{pmatrix}-4\\-2\end{pmatrix}, \quad -\nabla f_0\Big|_{\binom{1}{1}} = \begin{pmatrix}4\\2\end{pmatrix}$$

$$-\nabla f_0\Big|_{\binom{1}{1}} = \begin{pmatrix}4\\2\end{pmatrix} = 3 \times \underbrace{\begin{pmatrix}0\\2\end{pmatrix}}_{\nabla f_1\big|_{\binom{1}{1}}} + 4 \times \underbrace{\begin{pmatrix}1\\-1\end{pmatrix}}_{\nabla f_3\big|_{\binom{1}{1}}}$$

Stat. Cond

$\hat{\lambda}_1 = 3, \quad \hat{\lambda}_2 = 0, \quad \hat{\lambda}_3 = 4$

comp. sl.